Notes: Krugman (JPE, 1991)

Increasing Returns and Economic Geography

Contents

1 Big picture

- **Question.** Why and when does manufacturing concentrate geographically in one region, leaving other regions as an agricultural periphery?
- **Setting.** A two-region economy with agriculture tied to immobile land and manufacturing characterized by increasing returns, product differentiation, and transport costs.
- **Core mechanism.** Manufacturing firms want to locate near large markets to save transport costs, but the size of the market itself depends on where manufacturing workers and firms are already located. This creates circular causation.
- **Main theoretical result.** A core-periphery pattern can arise endogenously from **pecuniary externalities** — backward and forward linkages — once transport costs are low enough, economies of scale are strong enough, and manufacturing is large enough in expenditure.
- **Key comparative statics.** Regional divergence is more likely when transport costs are lower, when the expenditure share on manufactures is higher, and when economies of scale are stronger (equivalently, when the elasticity of substitution is lower).
- **Main intuition.** Three forces determine whether manufacturing disperses or agglomerates: the **home-market effect**, the **price-index effect**, and the **competition effect**. The first two push toward concentration; the last pushes toward dispersion.
- **Why this paper matters.** It gives a tractable formal model of economic geography using the Dixit-Stiglitz monopolistic-competition framework and shows that regional concentration can arise without invoking mysterious technological spillovers.

2 Data and measurement (key objects)

This is a **theory paper**; there is no empirical dataset. The "key objects" are the regions, factors, prices, and parameters that define the model and its numerical illustrations.

2.1 Economic units and sectors

- **Regions.** There are two regions, indexed by 1 and 2.
- **Sectors.** There are two sectors:
  - agriculture, which has constant returns and is tied to land;
  - manufacturing, which has increasing returns and can locate in either region.
- **Factors.** Following Krugman's simplifying assumption, each factor is specific to one sector:
  - peasants produce agriculture and are immobile across regions;
  - manufacturing workers produce manufactures and may migrate across regions.

2.2 Key parameters and normalizations

- **Manufacturing expenditure share** μ. Consumers always spend a share μ of expenditure on manufactures.
- **Elasticity of substitution** $\sigma > 1$. This governs substitution across manufacturing varieties and, in equilibrium, is inversely related to effective scale economies.
- **Transport parameter** $\tau \in (0, 1]$. If one unit of manufactures is shipped across regions, only a fraction τ arrives. Higher τ means lower transport costs.
- **Fixed and marginal labor requirements** α and β in manufacturing.
- **Normalization.** Krugman chooses units so that the total mass of manufacturing workers also equals μ:

$$L_1 + L_2 = \mu.$$

  This normalization makes the long-run wage of manufacturing workers equal to the peasant wage under symmetry.

2.3 Endogenous variables

- **Worker allocation:** L_1, L_2.
- **Number of manufacturing varieties by region:** n_1, n_2.
- **Nominal wages of manufacturing workers:** w_1, w_2.
- **Manufacturing price indices:** P_1, P_2.
- **Real wages:** ω_1, ω_2.
- **Regional incomes:** Y_1, Y_2.

2.4 Timing and equilibrium concepts

- **Short-run equilibrium.** The allocation of manufacturing workers across regions is taken as given.
- **Long-run equilibrium.** Workers migrate toward the region with the higher real wage.
- **Concentration test.** Instead of solving global dynamics everywhere, Krugman asks whether a fully concentrated allocation is an equilibrium — i.e., whether a single firm would want to "defect" to the peripheral region.

2.5 Numerical illustrations

- Figure 1 fixes $\sigma = 4$ and $\mu = 0.3$ and compares two values of transport costs, $\tau = 0.5$ and $\tau = 0.75$.
- Figure 2 studies the concentration condition as a function of τ for $\sigma = 4$ and $\mu = 0.3$.
- Figure 3 traces the concentration boundary in (μ, τ) space for $\sigma = 4$ and $\sigma = 10$.

3 Models and empirical specifications (all equations + interpretation)

3.1 Utility and expenditure shares

All individuals have utility

$$U = C_M^{\mu} C_A^{1-\mu}, \tag{1}$$

where C_A is agricultural consumption and C_M is the manufacturing aggregate.

The manufacturing aggregate is CES:

$$C_M = \left[\sum_{i=1}^{N} c_i^{(\sigma-1)/\sigma} \right]^{\sigma/(\sigma-1)}. \tag{2}$$

- **Interpretation.** Cobb-Douglas utility between agriculture and manufactures makes the manufacturing expenditure share constant at μ.
- **Interpretation of σ.** The higher is σ, the easier it is to substitute across varieties, and the weaker are equilibrium scale economies.

3.2 Factor supplies and production technology

Peasants are immobile and evenly split across the two regions. Manufacturing workers can move across regions, with

$$L_1 + L_2 = \mu. \tag{3}$$

Each manufactured variety is produced with fixed and marginal labor requirements:

$$L_{Mi} = \alpha + \beta x_i, \tag{4}$$

where x_i is output and L_{Mi} is labor used by firm i.

- **Interpretation.** Equation (4) is the increasing-returns core of the paper: there is a fixed cost α and constant marginal cost β.
- **Geography side.** Agriculture is tied to land, while manufacturing is footloose. Hence the spatial question concerns manufacturing only.

3.3 Transport costs and firm pricing

Agriculture is assumed costless to transport. Manufacturing trade costs are of Samuelson iceberg form: shipping one unit across regions delivers only $\tau < 1$ units.

With CES demand, each manufacturing firm faces elasticity σ and charges a constant markup over marginal cost. In region 1,

$$p_1 = \left(\frac{\sigma}{\sigma - 1} \right) \beta w_1, \tag{5}$$

and similarly in region 2. Hence relative prices satisfy

$$\frac{p_1}{p_2} = \frac{w_1}{w_2}. \tag{6}$$

Free entry drives profits to zero:

$$(p_1 - \beta w_1)x_1 = \alpha w_1, \tag{7}$$

which implies a common firm output level,

$$x_1 = x_2 = \frac{\alpha(\sigma - 1)}{\beta}. \tag{8}$$

Therefore the number of varieties is proportional to the number of manufacturing workers:

$$\frac{n_1}{n_2} = \frac{L_1}{L_2}. \tag{9}$$

- **Interpretation.** All firms are symmetric in equilibrium conditional on location, so location affects the number of firms rather than firm size.
- **Scale-economy interpretation.** Under zero profits, $\sigma/(\sigma - 1)$ is the ratio of marginal to average product of labor. A lower σ means stronger effective scale economies.

3.4 Short-run equilibrium: demand ratios and regional incomes

Define c_{11} as consumption in region 1 of a representative variety produced in region 1, and c_{12} as consumption in region 1 of a representative variety produced in region 2. Since imported goods bear iceberg costs, relative demand in region 1 is

$$\frac{c_{11}}{c_{12}} = \left(\frac{p_1\tau}{p_2}\right)^{-\sigma} = \left(\frac{w_1\tau}{w_2}\right)^{-\sigma}. \tag{10}$$

Let z_{11} be the ratio of region 1 expenditure on local manufactures to expenditure on imported manufactures. Then

$$z_{11} = \left(\frac{n_1}{n_2}\right)\left(\frac{p_1\tau}{p_2}\right)\left(\frac{c_{11}}{c_{12}}\right) = \left(\frac{L_1}{L_2}\right)\left(\frac{w_1\tau}{w_2}\right)^{-(\sigma-1)}. \tag{11}$$

Similarly, in region 2,

$$z_{12} = \left(\frac{L_1}{L_2}\right)\left(\frac{w_1}{w_2\tau}\right)^{-(\sigma-1)}. \tag{12}$$

Manufacturing wage income in region 1 equals manufacturing expenditure on region 1 goods from both regions:

$$w_1L_1 = \mu\left[\left(\frac{z_{11}}{1+z_{11}}\right)Y_1 + \left(\frac{z_{12}}{1+z_{12}}\right)Y_2\right], \tag{13}$$

and similarly for region 2,

$$w_2L_2 = \mu\left[\left(\frac{1}{1+z_{11}}\right)Y_1 + \left(\frac{1}{1+z_{12}}\right)Y_2\right]. \tag{14}$$

Regional income includes peasant income plus worker income:

$$Y_1 = \frac{1-\mu}{2} + w_1 L_1, \tag{15}$$

$$Y_2 = \frac{1-\mu}{2} + w_2 L_2. \tag{16}$$

- **Interpretation.** The wage in a region is shaped by two opposing forces. A larger region enjoys a **home-market effect** because local demand is larger, but firms there also face more **competition** for the local peasant market.
- **Marshallian short run.** The distribution of workers is fixed, so (11)–(16) determine nominal wages and incomes conditional on geography.

3.5 Long-run equilibrium: price indices and real wages

Let

$$f = \frac{L_1}{\mu},$$

be the share of manufacturing workers in region 1. Then the price index of manufactured goods for consumers in region 1 is

$$P_1 = \left[f w_1^{-(\sigma-1)} + (1-f)\left(\frac{w_2}{\tau}\right)^{-(\sigma-1)} \right]^{-1/(\sigma-1)}, \tag{17}$$

and for consumers in region 2,

$$P_2 = \left[f \left(\frac{w_1}{\tau}\right)^{-(\sigma-1)} + (1-f) w_2^{-(\sigma-1)} \right]^{-1/(\sigma-1)}. \tag{18}$$

Real wages are

$$\omega_1 = w_1 P_1^{-\mu}, \tag{19}$$

$$\omega_2 = w_2 P_2^{-\mu}. \tag{20}$$

- **Interpretation.** Even if nominal wages are the same, workers in the larger region face a lower price index for manufactured goods because more varieties are produced nearby. This is the **price-index effect**, and it pushes toward concentration.
- **Dynamic rule.** If ω_1/ω_2 rises with f, workers migrate into the already larger region and we get divergence. If it falls with f, the larger region loses workers and we get convergence.
- **Three forces.** The long-run geography is governed by the home-market effect, the competition effect, and the price-index effect.

3.6 Numerical characterization around the symmetric allocation

Krugman first studies the model numerically by evaluating ω_1/ω_2 as a function of f.

With $\sigma = 4$ and $\mu = 0.3$:

- when $\tau = 0.5$ (high transport costs), ω_1/ω_2 decreases with f, implying convergence;
- when $\tau = 0.75$ (low transport costs), ω_1/ω_2 increases with f, implying divergence.

- **Interpretation.** Lower transport costs strengthen the forces favoring agglomeration relative to the competition force.
- **Why numerical first.** The exact real-wage schedule is hard to solve analytically, so the paper turns next to a different analytical question.

3.7 Necessary condition for complete manufacturing concentration

Instead of asking whether the symmetric allocation is locally stable, Krugman asks whether a configuration in which all workers are concentrated in one region can be an equilibrium.

Suppose all manufacturing workers are in region 1. Then region 1 is the larger market, and because all manufacturing income accrues there,

$$\frac{Y_2}{Y_1} = \frac{1-\mu}{1+\mu}. \tag{21}$$

If n is the total number of manufacturing firms, each region 1 firm has value of sales

$$V_1 = \left(\frac{\mu}{n}\right)(Y_1 + Y_2). \tag{22}$$

Now ask whether a single "defecting" firm can profitably produce in region 2. To attract workers there, it must compensate them for the fact that almost all manufactured goods must be imported, so

$$\frac{w_2}{w_1} = \left(\frac{1}{\tau}\right)^{\mu} = \tau^{-\mu}. \tag{23}$$

Its value of sales is

$$V_2 = \left(\frac{\mu}{n}\right)\left[\left(\frac{w_2}{w_1\tau}\right)^{-(\sigma-1)} Y_1 + \left(\frac{w_2\tau}{w_1}\right)^{-(\sigma-1)} Y_2\right]. \tag{24}$$

So the ratio of defecting-firm sales to incumbent-firm sales is

$$\frac{V_2}{V_1} = \frac{1}{2}\tau^{\mu(\sigma-1)}\left[(1+\mu)\tau^{\sigma-1} + (1-\mu)\tau^{-(\sigma-1)}\right]. \tag{25}$$

Because fixed costs are also higher in region 2, the correct profitability condition is not $V_2/V_1 > 1$ but rather $V_2/V_1 > w_2/w_1 = \tau^{-\mu}$. Define

$$\nu = \frac{1}{2}\tau^{\mu\sigma}\left[(1+\mu)\tau^{\sigma-1} + (1-\mu)\tau^{-(\sigma-1)}\right]. \tag{26}$$

If $\nu < 1$, then a firm cannot profitably defect, so complete concentration of manufacturing in one region is an equilibrium. If $\nu > 1$, it is not.

- **Interpretation.** This is the paper's cleanest analytical device: rather than fully solving migration dynamics, test whether the periphery is unattractive enough that no firm wants to enter it.
- **Important nuance.** This is a **necessary condition for concentration**, not a full characterization of all stable equilibria. In principle, stable interior equilibria could also exist.

3.8 Comparative statics of the concentration condition

Differentiate ν with respect to the key parameters.

For the manufacturing expenditure share,

$$\frac{\partial \nu}{\partial \mu} = \nu\sigma \ln \tau + \frac{1}{2}\tau^{\mu\sigma}\left[\tau^{\sigma-1} - \tau^{-(\sigma-1)}\right] < 0. \tag{27}$$

A higher μ therefore makes concentration more likely.

For transport costs,

$$\frac{\partial \nu}{\partial \tau} = \frac{\mu\sigma\nu}{\tau} + \frac{\tau^{\mu\sigma}(\sigma-1)\left[(1+\mu)\tau^{\sigma-1} - (1-\mu)\tau^{-(\sigma-1)}\right]}{2\tau}. \tag{28}$$

At the relevant critical point, $\partial\nu/\partial\tau < 0$, so higher transport costs work against concentration.

For the elasticity of substitution,

$$\begin{aligned}\frac{\partial \nu}{\partial \sigma} &= \ln(\tau)\left\{\mu\nu + \frac{1}{2}\tau^{\mu\sigma}\left[(1+\mu)\tau^{\sigma-1} - (1-\mu)\tau^{-(\sigma-1)}\right]\right\} \\ &= \ln(\tau)\left(\frac{\tau}{\sigma}\right)\left(\frac{\partial \nu}{\partial \tau}\right).\end{aligned} \tag{29}$$

Since $\partial\nu/\partial\tau < 0$ at the boundary and $\ln(\tau) < 0$, it follows that $\partial\nu/\partial\sigma > 0$. Hence a higher σ — that is, weaker economies of scale — works against concentration.

The concentration boundary in (μ, τ) space therefore has

$$\frac{\partial \tau}{\partial \mu} = -\frac{\partial\nu/\partial\mu}{\partial\nu/\partial\tau} < 0,$$

and, holding μ fixed,

$$\frac{\partial \tau}{\partial \sigma} = -\frac{\partial\nu/\partial\sigma}{\partial\nu/\partial\tau} > 0.$$

- **Interpretation.** The parameter combinations that sustain agglomeration are exactly the intuitive ones: a larger manufacturing share, lower transport costs, and stronger scale economies.
- **Extreme case.** If $\sigma(1-\mu) < 1$, then concentration can be sustained no matter how high transport costs are, because manufacturing is large enough or scale economies are strong enough.

4 Identification and instruments (what makes the estimates credible)

Because this is a theory paper, there are no econometric instruments. The relevant "identification" issue is whether the mechanism is pinned down cleanly by the model assumptions and whether the comparative statics follow transparently from them.

4.1 What standard geography stories leave unresolved

- Traditional location stories often appeal to Marshallian technological spillovers, pooled labor markets, or specialized inputs.

- Those mechanisms are plausible, but in a formal model they are hard to discipline because the geographic reach of the externality is usually left unspecified.
- Purely competitive models also struggle to make pecuniary linkages matter for equilibrium selection or dynamics.

4.2 What pins down the new mechanism

- **Imperfect competition.** Under monopolistic competition, pecuniary externalities become real because firms price above marginal cost.
- **Increasing returns.** Fixed costs in manufacturing make location choice matter.
- **Mobile manufacturing labor.** Migration is the propagation device through which regional advantages become self-reinforcing.
- **Iceberg transport costs.** Distance enters the model in one clear way and can be directly linked to agglomeration forces.

4.3 Why the concentrated-equilibrium test is informative

- Full global stability analysis of ω_1/ω_2 is hard analytically.
- The defecting-firm condition compresses the central economic question into the scalar statistic ν.
- Once ν is written down, comparative statics with respect to μ, τ, and σ are transparent.
- This is why the paper can produce a clean analytical boundary even though the full migration dynamics are messy.

4.4 What does and does not identify the result

- The paper is **not** identified by any empirical shock or statistical design.
- It **is** disciplined by a tightly parameterized mechanism in which three structural forces map directly into whether regions converge or diverge.
- The key lesson is therefore conceptual rather than econometric: concentration is an endogenous equilibrium outcome, not an imposed assumption.

4.5 Relation to the literature

- Relative to Marshallian localization stories, Krugman shifts the emphasis from technological spillovers to **pecuniary demand and supply linkages**.
- Relative to new trade theory, the paper takes the Dixit-Stiglitz-Krugman monopolistic-competition machinery and puts it into space.
- Relative to development models of circular causation and linkage effects (Hirschman, Myrdal), the paper provides a formal and tractable microfoundation.

- Historically, this paper is one of the foundational statements of what later became the **new economic geography** literature.

5 Tables and figures

The paper has **three figures and no tables**. The figures are numerical illustrations of the model's stability and concentration conditions.

5.1 Figure 1: Real wage ratio and the stability of the symmetric allocation

Read:

- The figure plots ω_1/ω_2 against the share of workers in region 1, f.
- With $\sigma = 4$ and $\mu = 0.3$, the slope depends on transport costs.
- When $\tau = 0.5$ (high transport costs), the curve slopes downward: a larger region has *lower* relative real wages, so migration reverses and the economy converges.
- When $\tau = 0.75$ (low transport costs), the curve slopes upward: a larger region has *higher* relative real wages, so migration reinforces concentration and the economy diverges.

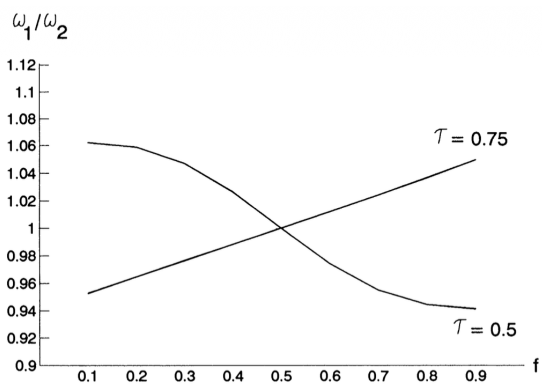

FIG. 1

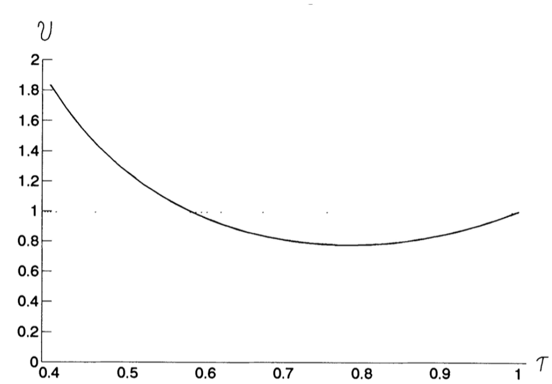

FIG. 2

5.2 Figure 2: The defecting-firm condition as transport costs vary

Read:

- The figure plots the statistic ν from equation (26) against τ for $\mu = 0.3$ and $\sigma = 4$.
- The horizontal benchmark is $\nu = 1$. If $\nu < 1$, a firm cannot profitably defect from the core to the periphery, so concentration is sustainable.
- At very low τ (high transport costs), $\nu > 1$, so the periphery is protected enough that defection is profitable and full concentration cannot be sustained.
- As transport costs fall, ν drops below 1, meaning concentration becomes a possible equilibrium; as $\tau \to 1$, location becomes irrelevant and $\nu \to 1$ again.

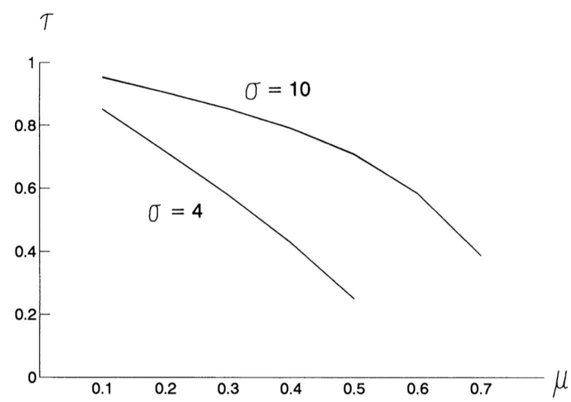

Fig. 3

5.3 Figure 3: Boundary in (μ, τ) space

Read:

- The figure traces the combinations of manufacturing share μ and transport parameter τ that separate concentration from nonconcentration.
- The boundary is downward sloping: if the manufacturing share is higher, concentration can arise even with higher transport costs.
- The curve for $\sigma = 10$ lies above the one for $\sigma = 4$, showing that weaker scale economies require more favorable transport/manufacturing conditions for concentration.
- Equivalently, stronger economies of scale (lower σ) make core-periphery outcomes easier to sustain.

5.4 What the figure set accomplishes overall

Read:

- Figure 1 gives the local stability intuition around symmetry.
- Figure 2 gives the global "defecting firm" intuition for complete concentration.
- Figure 3 summarizes the comparative statics in the two most important parameters, transport costs and manufacturing importance, for different strengths of scale economies.

6 Short summary

- **Conclusion.** Manufacturing concentration can arise endogenously from increasing returns, transport costs, and mobile workers, even without technological spillovers.
- **Main mechanism.** Backward and forward linkages create circular causation: firms want to locate where demand is large, and demand is large where firms and workers are located.
- **Main condition.** Agglomeration is more likely when transport costs are low, the manufacturing expenditure share is high, and economies of scale are strong.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that a simple monopolistic-competition model can generate a core-periphery pattern as an equilibrium outcome.
- It shows that the geography of manufacturing is not pinned down solely by exogenous land distribution; instead, endogenous market-size effects can overturn dispersion and produce concentration.
- It also shows that small parameter changes can trigger large qualitative changes in spatial equilibrium.

7.2 Economic intuition

- A manufacturing firm likes large markets because shipping is costly.
- A worker likes the region with more manufacturing because locally produced manufactures lower the cost of living.
- Once enough firms and workers are in one place, these two preferences reinforce each other.
- The only force pushing the other way is that firms in the larger region face more competition for the local agricultural market.
- Whether the economy agglomerates depends on which side wins.

7.3 Takeaway

This paper is best read as a foundational mechanism paper. Its main contribution is not empirical measurement but analytical clarity: it shows that **pecuniary externalities under increasing returns are enough to generate endogenous regional divergence**. That insight became the core of the new economic geography. The big lesson is that geography can be the outcome of general-equilibrium feedbacks rather than a passive reflection of fundamentals alone.